

Finance Economic and Management

Surplus = Profit

We don't want Loss

Company



Business

profit = Difference b/w
 expenses and incomes
 (cost price) (selling price)
 is one of the expenses (only one)
 is one of the incomes (few)

- preparatory
- partnership
- corporate

- public sector (Govt.)
- private sector €

24/07/26ECONOMICS

Marke Distribution : 10 + 10 + 10
 (F) (E) (M)

(Resource Allocation)

→ Unlimited Demand with scarce resources

Q. How ~~do~~ you to allocate the resources with maximum demand? ⇒ Economy guides it

- Micro Economics (science of individual decision making)
- Macro Economics (answers the most questions after micro economics)

→ aggregate behaviour
 of economy

GDP, ~~un~~ gross ~~and~~
 unemployment

Price Determination :

Which factors affect?

↳ Cost of Production

↳ Demand (low production cost but due to high demand price is high $\Rightarrow$ Coca Cola
high production cost but due to low demand price is low $\Rightarrow$)

Q. Difference between Desire & Demand?

Desire $\rightarrow$ wishing to have something

Demand $\rightarrow$ desire with purchasing capacity

Q. Primary drivers of Demand?

$$Q_x^d = f(P_x, M, P_y, \text{Taste \& Pref})$$

(Demand for quantity of commodity x)

$$P_x \propto \frac{1}{Q_x}$$

Price of substitute y

M: Income $\rightarrow$ influences the quantity to buy

Bang Valen effect

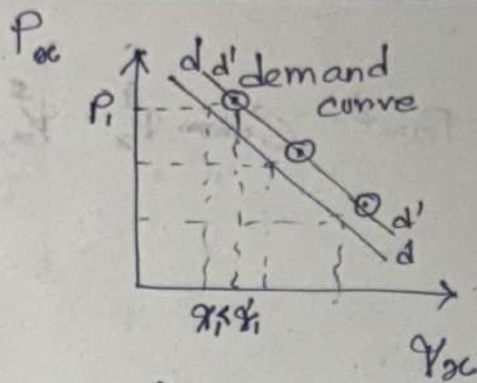
$\downarrow$

Then Due to trend some thing is bought

Law of Demand:

Quantity demanded of a product is inversely related to the price of the product, ceteris paribus.

$$Q_x \propto \frac{1}{P_x}, \text{ ceteris paribus}$$



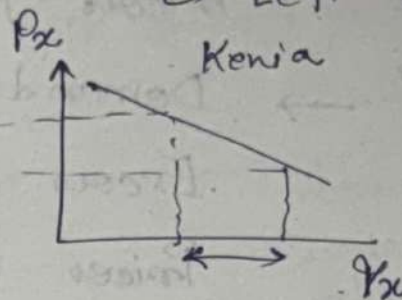
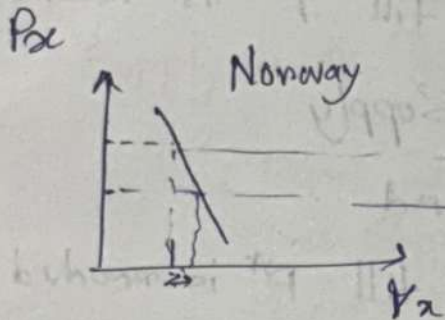
If M ~~changes~~ increases
demand ~~incre~~ of quantity
increases (at p same
price more quantity is
being demanded)

↳ Right shift parallelly

If M decreases

↳ Left shift parallelly

"Willingness to
~~produce~~ consumed
curve"



More the flat the curve is more
sensitive is the market consumers

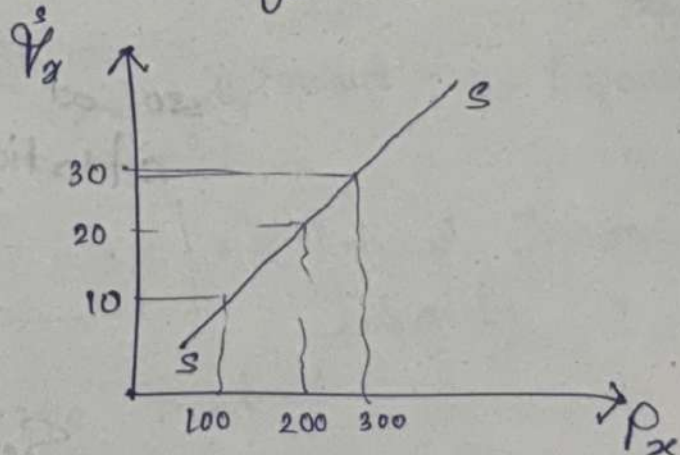
Small change in prices affect more
quantity of demand

Price incentive product → medical
drugs (Life Saving)

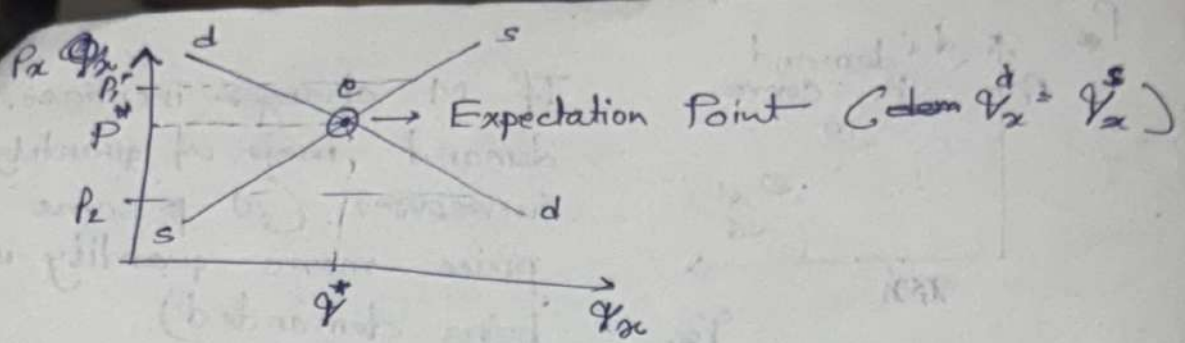
$$Y_x^s = f(P_x, P_{oc}, \text{Technology}, \gamma)$$

(Supply of
quantity of
commodity x)

$$Y_x^s \propto P_x$$

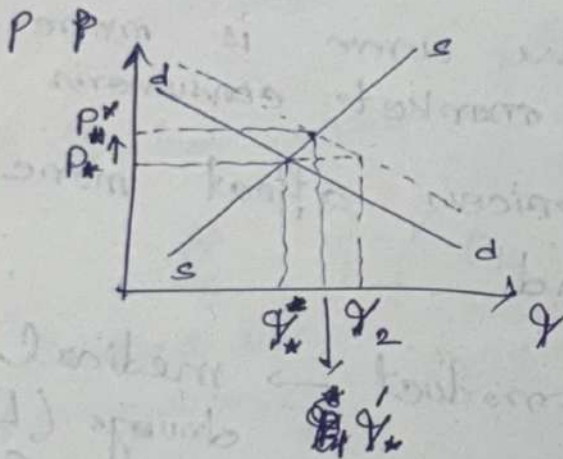


Willingness to ~~su~~ produce



At, $P_1 > P^*$ $\rightarrow$ Supply $>$ Demand
 Excess supply
 Prices fall till P^* is reached

At $P_2 < P^*$ $\rightarrow$ Demand $>$ Supply
 Excess demand
 Prices rise till P^* is reached



T & P increases $\rightarrow$ demand increases

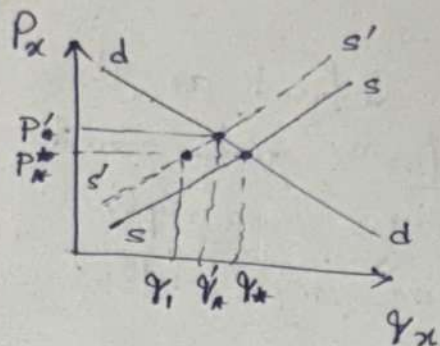
$\downarrow$
 parallel right shift of d-d curve

$\downarrow$
 so at P^* Excess demand situation occurs

$\downarrow$
 prices rise

$\downarrow$
 Some customers involve to buy

$\downarrow$
 $Q_2 \rightarrow Q_1$



During war supply decreases

↓
At P_* Excess demand situation

↓
price ~~decreases~~ increases

Economics

31/07/26

Supply $\downarrow \Rightarrow$ Price $\uparrow$ (ex. shifts) $\Rightarrow$ supply curve shift leftward

(More Demand)
(Less Supply)
Excess Demand

Adversarial
Demand
Shock

Microeconomics :

↳ How to increase the income level of an economy?

↳ How is income for an economy calculated?

↓
Natural Income Accounting → measure (calculating) for a whole ~~seem~~ economy.

In an economy,

Total economy $\equiv$ Total Production $\equiv$ Total Expenditure
(indirectly equal)

$\boxed{NI \equiv NP \equiv NE} \rightarrow$ National Income Identity

GDP: Gross Domestic Product

↳ total sum of all goods & services produced in the country

$\therefore GDP \uparrow \Rightarrow NP \uparrow \Rightarrow NI \uparrow$

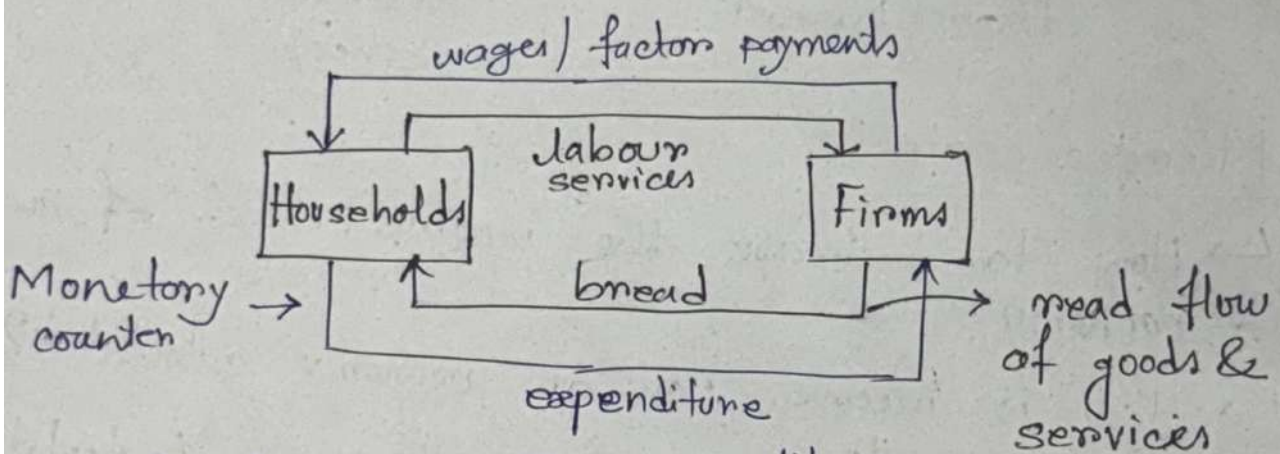
GDP:

Gross Domestic Product is defined as the market value of all final goods and services produced within the geographical / political boundaries of an economy within a specified period of time.

Small example of an economy, having 2

cities $\begin{cases} \rightarrow \text{Housing (households)} \\ \rightarrow \text{Farms} \end{cases}$

Circular Flow of Income Model :-



$$\begin{aligned} \text{Total Income} &= \text{Total Expenditure} \\ &= \text{Total bread Production} \end{aligned}$$

$\times \left\{ \begin{array}{l} \text{savings} \\ \text{external} \\ \text{investment} \end{array} \right.$

Assumptions:

- $\rightarrow$ ① Only one good is produced, i.e. bread
- $\rightarrow$ ② Only one factor of production is required to produce bread
- $\rightarrow$ ③ No leakages or injections in the form of savings or external investments rep. by households into farms

Here $NPI \equiv NP \equiv NE \Rightarrow$ Total value of bread produced $\equiv GDP$

Physically perceptible $\Rightarrow$ Good

" non- " $\Rightarrow$ Services

Valuation

07/08/26

higher per capita income $\Rightarrow$ higher welfare

National Income Accounting

How to measure?

$\hookrightarrow$ Sum total everybody's income

In an economy, National Income = National Expenditure = Total Goods & services produced (National Production)

$\hookrightarrow$ National Income Identity $\equiv GDP$

Circular flow of Income :

Simplified assumptions

① There are two sectors $\begin{cases} \rightarrow \text{Firms (Producers)} \\ \rightarrow \text{Household (Consumer)} \end{cases}$

② There is only one good produced : bread

③ " " " " factor of production : labour

④ No leakage or injections

GDP

Some key terms

- ↳ market value (sold @ price)
- ↳ final goods (that is consumed by final consumer)
- ↳ services
- ↳ specified geographical / political area (country, state, district)

↳ time

C-A

20 Laptop

10 units wheat

C-B

30 units of cloth

10 units AC

$$\begin{array}{l|l} \text{Price} \rightarrow P_1 & P_2 \dots P_n \\ \text{Quantity} \rightarrow Q_1 & Q_2 \dots Q_n \end{array} \quad \left| \quad P_1 Q_1 + P_2 Q_2 + \dots + P_n Q_n = \text{Total market value} \right.$$

$$\therefore \text{GDP} = \sum_{i=1}^n P_i Q_i$$

P_i : Price of i^{th} good produced

Q_i : quantity of i^{th} " " "

Feminist Economist $\rightarrow$ also count the unpaid labour given by women at our home

$\downarrow$
If accounted GDP will rise

How to increase GDP?

↳ Increase P

↳ " Q

↳ " $P \& Q$ both

↳ " n (num of products produced)

Most unethical way $\rightarrow$ increasing price
 $\downarrow$
inflation

ideal way $\rightarrow Y_i \uparrow, n \uparrow, (P_i \uparrow Y_i \uparrow)$

Increase of Y_i
outweighs the increase
of P_i

GDP defined here $\rightarrow$ Nominal GDP
(Based on prices of
current market)

C-X

Year 2020
 $P_a = 20/\text{unit}$ $Y_a = 30$ units
 $P_o = 10/\text{unit}$ $Y_o = 20$ "

$$GDP_N = (20 \times 30) + (10 \times 20)$$
$$= 800 \text{ rupees}$$

2025
 $P_a = 30/\text{unit}$ $Y_a = 40$
 $P_o = 25/\text{unit}$ $Y_o = 20$

$$GDP_N = (30 \times \overset{40}{\cancel{25}}) + (\overset{25}{\cancel{40}} \times 20)$$
$$= 1700 \text{ Rupees}$$

But is it the true reflection of increase
in of Economical activity?

$\rightarrow$ No (Apple prod increases $\rightarrow 33\%$)
(GDP increases $\rightarrow > 200\%$)

We set a base year (ie. fixed price) and
calculate price $\Rightarrow GDP_{real}$.

Let Base year = 2020

$$GDP_{real} \text{ of } 2025 = 20 \times 30 + 20 \times 10$$
$$= 1000 \text{ Rs}$$

Income :

$$\text{Increase} = \frac{1000 - 800}{800} \times 100\% = 25\%$$

$\rightarrow$ Consumption Expenditure.

$\rightarrow$ Investment "

$\rightarrow$ Govt "

$\rightarrow$ Net Exports "

Economics

$$GDP = NE = C + I + G + NX$$

If a company produces in India, profits are appropriated abroad, not part of GDP

what is the total production in India

what is the total income of people residing in India?

what is the total income of Indians?

GDP is independent of who is ~~the~~ doing that. Different

Nvidia → Profits are not profit of Indian citizens

does not part of GDP

Does TCS at San Francisco affect GDP of India? → no, not India

Gross National Product (GNP) = GDP +
factor income from abroad
- factor payments to abroad

$$= GDP + \text{net factor income from abroad}$$

$$GNP = GDP + NFIA$$

Net Factor Income from abroad +ve

$$GNP > GDP$$

GDP > GNP in India currently

we are paying more to foreign companies [As Indians are going out & working there]

Export is a component of GDP | export $\uparrow$ GDP $\uparrow$

China manufacturing mainly

$$GDP > GNP$$

Pakistan

$$GNP > GDP$$

Remittance based: works outside and send to home

GNP & GDP $\rightarrow$ Calculated on market prices

factor cost $\rightarrow$ cost of production

$\rightarrow$ also includes profits

$\neq$ ID $\rightarrow$ all included but

govt. will come & ask for tax

There is market distortion for taxes

subsidy at time of production (govt)

$\rightarrow$ $\downarrow$ price to make it affordable to the target customers.

$$GDP_{\text{Factor Cost}} = GDP_{MP} - \text{taxes} + \text{subsidies}$$

$$= GDP_{MP} - (\text{indirect taxes} - \text{subsidies})$$

$$= GDP_{MP} - \text{net indirect taxes}$$

$$GNP_{FC} = GNP_{MP} - \text{net indirect taxes}$$

$\rightarrow$ in order to arrive at the true economy activity

Depreciation: losing value of something

worth of depreciation: repair + (to keep it working)

$$NDP_{PC} = GDP - \text{depreciation}$$

(Net Domestic)

National Income = $NNP_{FC} = GNP_{FC} - \text{depreciation}$

How to increase GDP & National income?

R&D, Training, Startup $\rightarrow$ long gestation period

~~Recession~~ Recession, Recovery, Boom, Bust $\downarrow$
Economic waves

1930

The great dip

depression: GDP very low, high unemployment

$\rightarrow$ Hands off: no govt intervention, market will take care & correct

Laissez faire

$\hookrightarrow$ There is an invisible hand that will handle everything

$\rightarrow$ Another: in the long run but we will be dead

Govt: intervention leads to rise in GDP how $\rightarrow$ reset module

simple calculation model

You need to revive the demand in order to rescue to the falling economy

Expenditure

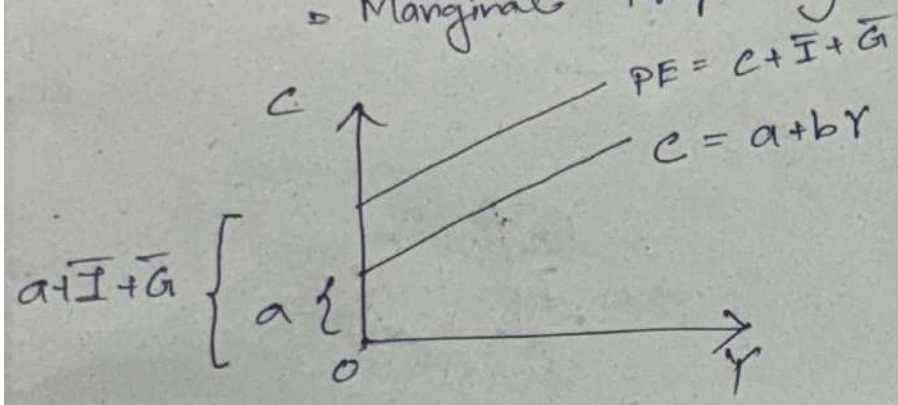
$$Y = C + \bar{I} + \bar{G}$$

$\downarrow$ $\downarrow$ $\downarrow$
 Income Consumer Income Govt
 Expenditure Investment

$$C = f(Y) = a + bY$$

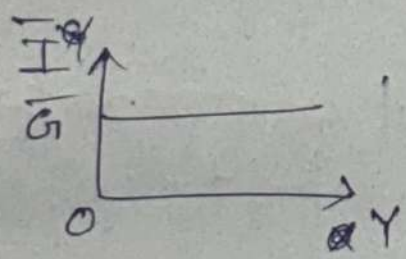
Even if $Y=0$, $C=a$ (subsistence level of consumption expenditure) > 0

b = rate at which consumption expenditure increases with the income $\in [0, 1]$
 = Marginal Propensity to Consume

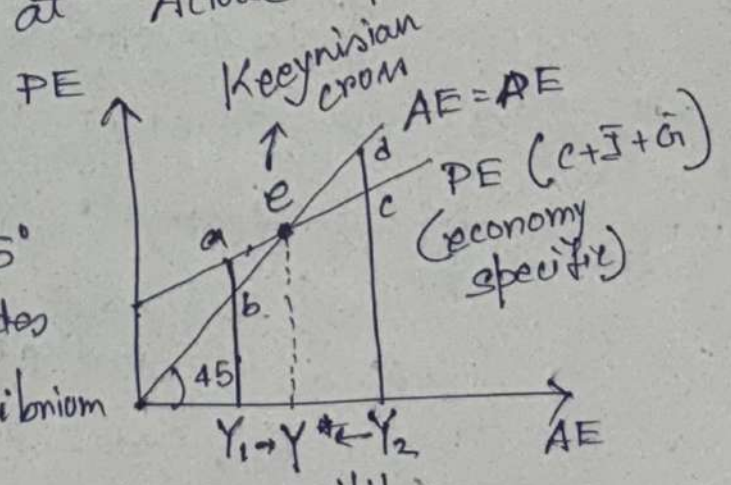


Plan Expenditure (PE) = $C + \bar{I} + \bar{G}$ → Constant

III Economy reaches to equilibrium, at Actual Expenditure = PE



all points on the 45° line are candidates for, Keynesian Equilibrium



$e \rightarrow$ stable equilibrium
 (if we deviate the variable it will come back at the same pos)
 (equilibrium level of income)

At income level Y_1 ,

$$PE = aY_1$$

$$(Production) AE = bY_1$$

Diagrammatically it is evident that,
 $aY_1 > bY_1$

$$\Rightarrow PE > AE$$

Diagrammatically

$\Rightarrow$ $\overset{\text{Total}}{\text{Demand}} > \overset{\text{Total}}{\text{Actual Production}}$

leads to $\Rightarrow$ Unanticipated Decumulation of Inventory

Firms reacts $\Rightarrow$ Production increases

Production level $Y_1 \rightarrow Y_2$ where $PE = AE$
increases

Expenditure

$$= \text{Consumption} + \text{Investment} + \text{Govt} + \text{Export}$$

$\rightarrow$ Durable Good
(have a considerable life span)

$\rightarrow$ Non-durable Good
(very short life span)

$\rightarrow$ Expenditure on Services

$\rightarrow$ Business Fixed Investment

$\rightarrow$ Inventory Investment

$\rightarrow$ Residential Investment
(used to buy houses)

Investment $\Rightarrow$ Change in Capital Stock of Goods

$\rightarrow$ Blast furnaces

Chairs & Tables for office

in Steel making industry

Negative Investment $\Rightarrow$ Disinvestment

Inventory $\Rightarrow$ Reserved goods

(To meet variable demand in market)

Require for adjustment

At level of Y_2 ,

$$PE = C_{Y_2}$$

$$AE = a + bY_2$$

Diagrammatically,

$$C_{Y_2} < dY_2$$

(Demand) $PE < AE$ (Production)

Unanticipated Accumulation of Inventory

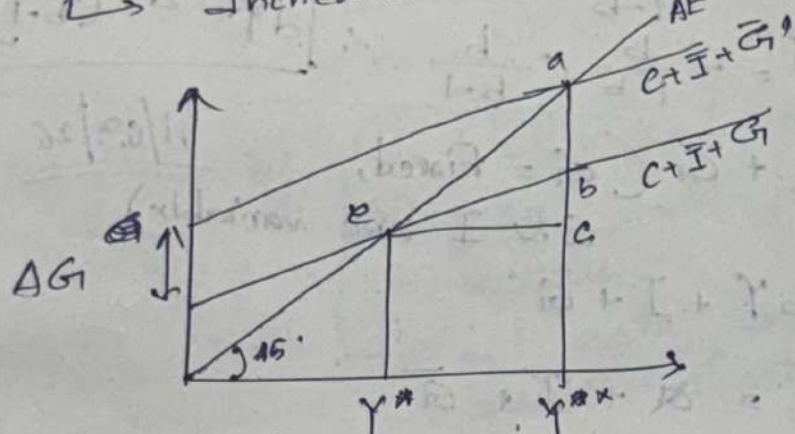
Production decrease

$$(Y_2 \xrightarrow{\text{Decreases}} Y^*)$$

How Increase the Y^* permanently rightward?

→ Increasing $a, b, \bar{I}$ (Not feasible in depressed economy)
($C = a + bY$)

→ Increase G (Govt Expend.)



$$\Delta G = ab$$

$$\Delta Y = ac$$

$\Delta Y > \Delta G$ (Diagrammatically)

$$Y = C + \bar{I} + G$$

$$Y = a + bY + \bar{I} + G$$

$$\Rightarrow Y(1-b) = a + \bar{I} + G$$

$$\Rightarrow Y = \frac{a}{1-b} + \frac{\bar{I}}{1-b} + \frac{G}{1-b}$$

$$\frac{dY}{dG} = \frac{1}{1-b}$$

$$\Rightarrow \left[dY = \frac{dG}{(1-b)} \right]$$

Govt Expenditure
Multiplier

HW

$$Y = C(Y - T) + \bar{I} + \bar{G}$$

lump sum Tax
What will be the resulting change in income Y in case of ~~tax~~ changing Tax T .

$$Y = (a + bY)(Y - T) + \bar{I} + \bar{G}$$

$$\Rightarrow Y = aY + bY^2 - aT - bYT + \bar{I} + \bar{G}$$

$$\Rightarrow \text{cf } (Y) = a + bY$$

$\therefore$ In case lump-sum tax, ~~cf~~ $(Y - T) = a + b(Y - T)$

$$\therefore Y = a + b(Y - T) + \bar{I} + \bar{G}$$

$$\Rightarrow Y(1 - b) = a - bT + \bar{I} + \bar{G}$$

$$\Rightarrow Y = a - \frac{bT}{1-b} + \bar{I} + \bar{G} \quad \boxed{dY = dT \left(\frac{b}{b-1} \right)}$$

$$\Rightarrow \frac{dY}{dT} = - \frac{b}{1-b} = \frac{b}{b-1}$$

Let, $Y = C + I + \bar{G}$ ($\bar{G}$ = Fixed, C & I are variable) 11/09/26

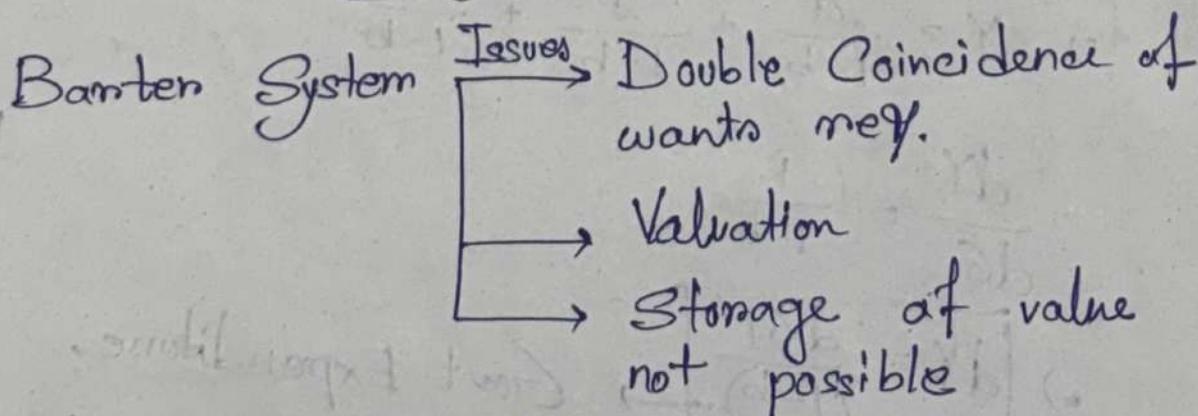
$$\Rightarrow Y = a + bY + I + \bar{G}$$

$$\Rightarrow Y(1 - b) = a + I + \bar{G}$$

$$\Rightarrow Y = \frac{a}{1-b} + \frac{I}{1-b} + \frac{\bar{G}}{1-b}$$

$$\frac{dY}{dI} = \frac{1}{1-b} \text{ (Investment Multiplier)}$$

Moneytony System



Solved by Money $\Rightarrow$ Medium of Exchange

Not the best way to store value because of inflation, but 2nd in line $\leftrightarrow$ Money

- Before Money, shells (kaudi's) were used as medium of exchange. But it was perishable.
- Thus came metallic money, mostly in gold. Due to being high value and possibility of counterfeiting its usage was difficult.
- Then Govt. issued certificates in exchange for gold.
 $\swarrow$ Promissory Notes
 Eventually currently delinked from gold.
 (notes)

Functions of Money

- ① Universal medium of exchange
- ② It serves as an unit of account
- ③ It acts as a store of wealth (not perfect)

Monetary Advantages :

Savings Account $\rightarrow$ Demand Deposits

$$\text{Total amount of money in circulation (M}_1\text{)} = \text{Currency in Circulation (CC)} + \text{Demand Deposit in comm. banks (DD)}$$

2nd def $M_2 = M_1 + \text{Savings deposits in Post Offices, Savings Banks}$

3rd def $M_3 = M_1 + \text{term deposits with Commercial Banks}$

4th def $M_4 = M_3 + \text{all deposits with P.O. Savings Banks}$

$M_1 \Rightarrow$ Most Liquid [$M_1 > M_2 > M_3 > M_4$]
 $M_1 \rightarrow$ Narrow Money
 $M_3 \rightarrow$ Broad Money

← Liquidity Inc.

Demands for Money

Reasons	①	②	③
Transaction demand for money			
Precautionary	"	"	"
Speculative	"	"	"

↳ when capital loss is speculated

Fiscal Policy : Related to Govt Expd. and Taxes

↳ Expansionary
 ↳ Contractionary

Monetary Policy → Related to Money Supply and demand by the RBI (Central Bank)

Banks → Lender / Bank of Last Resort

↳ Central Bank → Regulatory
 ↳ Commercial Bank for commercial bank
 ↳ Profit Oriented

Central Banks → price stability

↳ Growth of economy central
 ↳ Bankers of Banks

Public cannot open account in Central Bank but in Commercial Banks.

Q. What is management?

① Identify the problem (Finding resources available)

② Planning

③ Decision Making

④ Organizing (Distributing work, money...)

⑤ Leading

⑥ Motivating, ⑦ Controlling

Doing efficiently & effectively $\Rightarrow$ Cost Minimizing & Output Maximizing

$\Rightarrow$ Getting things done by other people

5 Functions of Management

$\hookrightarrow$ Planning (How you are going to work!)
Examining the future, what needs to be done and prepare plan)

$\hookrightarrow$ Organizing (Divide the resource, creating tasks & projects and dividing them)

$\hookrightarrow$ Commanding (Giving instructions to subordinate to get result)

$\hookrightarrow$ Coordinating (Harmonizing the work)

$\hookrightarrow$ Controlling (Verification whether the work is going according to command & plan, if there is any gap b/w desire & outcome)

Characteristics of Management:

- (i) Goal Oriented
- (ii) Universal Activity
- (iii) Continuous Process
- (iv) Dynamic Activity
- (v) Group Activity
- (vi) Distinct Process
- (vii) Profession (Requires sp. training and governed by ethics & principles)
- (viii) Multi-disciplinary
- (ix) Science & as well as an Art

↓
Follows some definite principles & common methodology

Requires creativity, intuition, response at time

(x) Intangible

Vertical Rank:

- (1) Top Level Management
(CEO, BoDin., M.Ds)
(Gives Goals, Visions, Decision maker)
- (2) Middle Level Management
(Implements the mission, visions or goals and general company goals)
- (3) Lower / Bottom Level Management
(Reporting authority, do the day to day activity)

What makes a team successful? → Motivation
↓ comes from

Type of People in Group:

- (1) Motivated & efficient
- (2) Not Motivated but efficient
- (3) Motivated but not efficient
- (4) Neither Motivated & efficient

Incentive
Respect
praise

Google's Project

1 Aristotot Aristotot

Styles of Management:

- ① Autocratic / Authoritarian / Directive
- ② Democratic / Participative / Inclusive
- ③ Laissez-faire / Hands off / Delegative

① Autocratic

- Manager / Leader makes decisions without feedback from subordinate
- Single Leadership
- Strict Control ①
- Swift Decision Making ②
- Clean Hierarchy ③
- Independent Leadership ④
- No input from team ⑤

Effective during emergencies, → in military

② Democratic

- Decision making shared across leaders and team members
- ① Collaborative decision making
- ② Flat hierarchy
- ③ ~~Low~~ empowerment

③ Laissez-faire

- Let it be
- Leaders provide very minimal info to members and allow them high level of autonomy to approach the task

- members are very skilled & motivated
- minimal direct supervision
- very little hierarchical structure
- encourages creativity
- more autonomy

Case Studies :

Google's Aristotle Project :

Research on what makes a team successful ?

- 108 teams worked with Google across countries
- Most Imp. is Psychological Safety of the employees
- Most star performers do not make a good team
- Employees are ~~not~~ encouraged to take risk ~~at~~ without being judged

South-west Airlines Management Style :

- Focus on employee empowerment
 - ↳ improvise customer exp. without waiting for a hierarchical exchange

- Increased employee morale

Toyota's Lean Manufacturing System (Study)